

Holly Kular

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SUMMARY

PhD-level machine learning scientist specializing in deep learning and neural network architectures. 5+ years of experience designing and training custom models, optimizing ML pipelines, and analyzing large-scale real-world datasets. Proficient in PyTorch, TensorFlow, and Python data processing libraries.

SKILLS

Deep learning: PyTorch, TensorFlow, neural networks (RNNs, SNNs, CNNs), image processing (MATLAB), model capacity analysis, network lesioning & interpretability, dimensionality reduction (PCA, ICA)

ML & Data Science: Scikit-learn, Python (Pandas, NumPy), R (tidyverse), regression (linear, regularized), classification (logistic regression, SVM, k-NN, naïve Bayes), time-series analysis & forecasting, SQL

Statistical Analysis: Bayesian modeling & inference, mixed-effects models, ANOVA, permutation tests & hypothesis testing, power simulations

Tools & Infrastructure: Git, Docker, distributed GPU computing (CUDA/PyTorch), MATLAB, R

EXPERIENCE

Graduate Scientist, University of California, San Diego

2021-Present

- Designed, trained, and evaluated custom recurrent neural networks for continuous visual decision-making classification tasks; conducted systematic misclassification analysis and edge-case investigation to validate model robustness.
- Modernized legacy TensorFlow 1.0 codebase by translating to PyTorch 2.5+, enabling compatibility with modern CUDA/GPU hardware and enabling benchmarking across model versions.
- Engineered and preprocessed features from large-scale human behavioral datasets (over 200,000 responses), applying cross-validation and both parametric and non-parametric statistical benchmarking to ensure reproducibility.
- Performed statistical analyses (GLM, mixed-effects models, ANOVA, permutation tests, Bayesian inference) and generated data visualizations to communicate findings to technical and non-technical audiences.
- Accelerated neural network evaluation and classification performance by 20x through distributed GPU use, demonstrating experience optimizing ML pipeline performance at scale.
- Conducted systematic neural network capacity analyses using entropy-based unit analysis to identify minimal sufficient architectures.
- Performed lesion studies to characterize learned representations and hidden state dynamics – informing efficient model design

Research Assistant, Stanford University

2019 - 2021

- Containerized a multi-site analysis pipeline (Docker) for a large, high-dimensional clinical imaging and health dataset, streamlining data cleaning, preprocessing, and reproducibility – contributing to 4 peer-reviewed publications.
- Coordinated data and project updates across teams on a multi-site grant, translating technical findings into accessible summaries for diverse stakeholders.

SELECTED PROJECTS

Deep Learning Models of Visual Decision Strategies

Investigated learned representations through strategic lesioning and entropy-based capacity analysis, identifying the minimal architecture sufficient for the task.

Visual Memory Error Prediction: modeling human behavioral data

What causes visual memory errors? I developed a model to predict memory errors in visual decision-making tasks leveraging large datasets of human behavior to identify parameters that influence memory performance.

EDUCATION

University of California, San Diego, PhD in Computational Neuroscience

Expected December 2026

University of California, Los Angeles, BS in Neuroscience

2019